A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Energy conservation: transport, buildings, industry - Set 1

## Mark scheme

AQA spec 3.3.4.2; Energy resources; textbook pp. 225-235

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **Calculation**<br>1 mark for the correct answer; working does not have to be shown.<br>**1** (2.9 - 0.7) / 2.9 x 100 = 75.9% (accept 76%)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (%) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 1 |
| **01.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Although the larger vehicle uses more energy in total, it carries many more people / a larger load<br>**1** So the energy per passenger (or per tonne of goods) is lower<br>– steel wheels on rails also have low rolling resistance<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |



Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.3** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** Flexibility of routes / destinations and door-to-door convenience<br>**1** Time / speed and reliability where public transport is infrequent<br>**1** Low fuel tax / cheap fuel<br>**1** Carrying goods<br>– comfort<br>– status<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **reliability** → repeatable / gives consistent results when repeated<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument | 3 |
| **01.4** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey<br>**1** So more journeys shift from cars to bulk transport / cycling<br>**1** ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely<br>**1** Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1. | |
| **02.1** | **2 marks per point**<br>2 marks for each of the 3 points below; award 1 mark where a point is only partly made.<br>**2** Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed<br>– hydrodynamics for ships<br>**2** Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption<br>**2** Conventional friction brakes convert kinetic energy to waste heat<br>– regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 6 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Pneumatic tyres deform as they rotate<br>– the continual change in shape and air movement inside generates frictional heat (more if under-inflated)<br>**1** Steel wheels do not deform so rolling resistance is very low<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **02.3** | **Any 2 of 4**<br>1 mark each for any 2 of the 4 points below, in any order, maximum 2.<br>**1** More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel<br>**1** Better temperature control (thermostatic fan)<br>**1** Electronic ignition control / regular servicing to keep spark timing correct<br>**1** Turbochargers supplying more oxygen<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1. | |
| **03.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** The energy used to extract raw materials, manufacture the components and assemble the vehicle (for a typical car about 1.5 years of its fuel use)<br>**1** Reduced by using recycled materials (eg recycled aluminium needs 1/20 of the energy of new)<br>**1** Lighter designs using less material<br>– longer vehicle life spreading the embodied energy<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **embodied energy** → all the energy used to make and transport a product | 3 |
| **03.2** | **Any 4 of 5**<br>1 mark each for any 4 of the 5 points below, in any order, maximum 4.<br>**1** Use of recyclable materials wherever possible<br>**1** Easy identification of components and their composition (code stamps)<br>**1** Easy dismantling and separation of components / materials<br>**1** Reusable components for new vehicles<br>**1** Compostable materials for parts that cannot be recycled<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** At low speed a large proportion of the energy just runs the engine rather than moving the car<br>**1** At high speed energy is wasted overcoming air resistance (which rises with the square of speed)<br>– optimum for many cars about 56 mph<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **03.4** | 1 mark for the point below.<br>**1** Automatic stop / start systems cutting the engine when stationary<br>– smooth driving in higher gears avoiding sudden braking / acceleration<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 227-228; AQA spec 3.3.4.2.1. | |
| **04.1** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1 Single:** 5.8 x 20 x 12 = 1392 W<br>**1 Argon:** 1.6 x 20 x 12 = 384 W<br>**1** Reduction = 1008 W (accept 1010)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The converse of any point, ie the same comparison written the other way round.<br>• The value alone - the unit (W) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 3 |
| **04.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** A wider gap of trapped gas reduces conduction because gas is a poor conductor<br>**1** If the gap is too wide convection currents start in the gas, carrying heat across<br>– the optimum depends on the gas and window size<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

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| Question | Marking guidance | Marks |
|---|---|---|
| **04.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Low-e coating reflects long-wavelength infrared back into the room, reducing radiation losses<br>**1** Argon / krypton / xenon have lower thermal conductivity than air so less conduction across the gap<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **infrared** → heat radiation / long-wave radiation re-emitted by the Earth | 2 |
| **04.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Panes must be held apart by small spacers which cannot be long, so the gap is only about 0.2 mm<br>**1** Used where there is not room for thick gas-filled multiple glazing (eg historic buildings)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **04.5** | 1 mark for the point below.<br>**1** Each extra layer saves less energy (diminishing returns) while cost rises, so beyond the optimum the cost of insulation exceeds the value of energy saved<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-231; AQA spec 3.3.4.2.2. | |
| **05.1** | **2 marks per point**<br>2 marks for each of the 3 points below; award 1 mark where a point is only partly made.<br>**2 Orientation:** south-facing windows (northern hemisphere) maximise passive solar gains<br>– rooms needing warmth on the sunny side<br>– fewer / smaller windows on the north side where losses are greatest<br>– solar screens / brise soleil prevent summer overheating<br>**2** Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost<br>**2 Earth sheltering:** the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 6 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **05.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy<br>**1** Limecrete has about half the embodied energy of concrete<br>– rammed earth, straw bales and lime mortar are lower still<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **embodied energy** → all the energy used to make and transport a product | 2 |
| **05.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area<br>**1** Heat transfers from the outgoing to the incoming air so fresh air enters already warmed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2. | |
| **06.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Occupancy sensors detect infrared / movement and switch lights or appliances off when no one is in the room<br>**1** Programmable thermostats set different temperatures for different rooms and times so unused rooms / periods are not heated<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **infrared** → heat radiation / long-wave radiation re-emitted by the Earth | 2 |
| **06.2** | **Marks shared**<br>Up to 3 marks in total - share them between the 2 points below according to how completely each one is made.<br>**up to 2** LEDs convert a much larger proportion of electricity to light rather than heat<br>**up to 2** Washing machines with cold / low-temperature cycles and fast spins (less drying)<br>– dishwashers using less water so less heating<br>– cookers with double / triple glazed doors<br>– fridges with linear compressors<br>– LED screens using half the power of plasma<br>– germanium transistors<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |

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| Question | Marking guidance | Marks |
|---|---|---|
| **06.3** | **Any 3 of 5**<br>1 mark each for any 3 of the 5 points below, in any order, maximum 3.<br>**1** Turning off lights / appliances rather than standby<br>**1** Turning thermostats and radiator valves down / off in unused rooms<br>**1** Only heating the water needed (kettle)<br>**1** Showering instead of bathing<br>– eco-cycles<br>**1** Choosing smaller vehicles / less travel<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. | 3 |
| **06.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Stored hot water loses heat continuously through the tank walls and pipes<br>**1** Heating as needed (or just before) avoids these standing losses<br>– if storage is needed the tank should be well insulated<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 232-233; AQA spec 3.3.4.2.2. | |
| **07.1** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1 Conventional:** 40 / 100 = 40%<br>**1 CHP:** (35 + 45) / 100 = 80%<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (%) is already printed on the answer line.<br>• **CHP** → using the waste heat from generation for heating<br>**Do not credit**<br>• A bare wrong answer with no working. | 2 |
| **07.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Extracting less energy as electricity leaves the cooling / condenser water at a higher temperature<br>**1** Hotter water is more useful for district heating of homes, fish farms, greenhouses<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **CHP** → using the waste heat from generation for heating | 2 |

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| Question | Marking guidance | Marks |
|---|---|---|
| **07.3** | 1 mark for the point below.<br>**1** Hot water loses heat and needs pumping energy over distance<br>– heat cannot be transported as far as electricity<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **CHP** → using the waste heat from generation for heating | 1 |
| **07.4** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Long narrow pipes give a large surface area and copper is a good thermal conductor so heat transfers quickly<br>**1** Hot and cold fluids flowing in opposite directions maintain a temperature gradient along the whole length so more heat is transferred<br>**1** Slower flow allows more time for heat to transfer<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **07.5** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** One large tank has a lower surface area : volume ratio than many small ones<br>**1** A sphere has the smallest surface area for any volume<br>– less surface means less heat loss (with insulation of low thermal conductivity)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 233-235; AQA spec 3.3.4.2.3. | |
| **08.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Molten iron from the blast furnace is converted directly to steel on the same site<br>**1** So it does not cool and need re-melting<br>– transport energy between sites is also avoided<br>– waste heat from one process used by another (eg Kalundborg)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

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| Question | Marking guidance | Marks |
|---|---|---|
| **08.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Collecting and processing small dispersed quantities can use more energy in transport than is saved<br>**1** Mass production and large-scale mining have economies of scale<br>– some materials (glass) take more energy to make than plastic but can be refilled many times<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **recycling** → reprocessing a material to make something new<br>**Do not credit**<br>• 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new | 2 |
| **08.3** | **2 marks per point**<br>2 marks for each of the 2 points below; award 1 mark where a point is only partly made.<br>**2** Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared)<br>– overhead cables cheaper than underground which need cooling<br>**2** ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **pumped storage** → water pumped uphill when demand is low and released to generate at peak | 4 |
| **08.4** | **Any 2 of 4**<br>1 mark each for any 2 of the 4 points below, in any order, maximum 2.<br>**1** Proximity to the grid / demand to reduce transmission losses and cable costs<br>**1** Access to cooling water for thermal stations<br>**1** Resource availability (wind, tidal range, fuel deliveries)<br>**1** Land availability and planning / public acceptance<br>– stable geology<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3. | |

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